

Comprehensive Production Audit

Senior Staff Architectural, Performance, Security & Recruiter Assessment

TARGET:	EVALUATION STANDARDS:	READINESS STATUS:
Lakshya Agarwal (lakshya.uk) Stack: Next.js 14, React 18, D3	OpenAI, Google, Meta Staff Criteria WCAG 2.2 AAA & OWASP Top 10	NOT PRODUCTION READY 4 Critical Deployment Blockers

[!] CRITICAL AUDIT VERDICT: DEPLOYMENT CURRENTLY BLOCKED
The portfolio features exceptional visual ambition, but contains critical flaws (broken 404 resume download, unthrottled 22MB frame bursts, duplicate window spawning, zero security headers, and WCAG audio/cursor violations).

1. Domain Breakdown & Evaluation Scores

DOMAIN	SCORE	BENCHMARK STANDARD	PRIMARY FINDINGS
Architecture	6.5 / 10	Next.js 14 Server/Client	Monolithic 'use client'; TopoJSON synchronous bu
Code Quality	7.0 / 10	TypeScript rigor, Clean	Canvas width/height set on scroll; 300 state chu
Security	4.0 / 10	OWASP Top 10, CSP, API G	Zero HTTP security headers; un-rate-limited mock
Performance	5.5 / 10	Core Web Vitals, LCP < 1	22MB unthrottled WebP burst; continuous 60fps of
Accessibility	3.5 / 10	WCAG 2.2 AAA, Keyboard N	Global cursor:none !important; unconsented audio
UI / UX Polish	8.5 / 10	Visual hierarchy, Polish	High creative craft; watermark comment in source
Animations	8.0 / 10	Framer Motion physics, G	Fluid physics; missing prefers-reduced-motion me
Visual Identity	9.0 / 10	Aesthetic elegance, Typo	Outstanding custom cyberpunk theme; cohesive dar
Maintainability	6.0 / 10	Component modularity, Te	Files exceed 900+ lines; zero automated unit or
Recruiter Appeal	6.0 / 10	FAANG / OpenAI Hiring Cr	Broken Resume PDF (404); WhiteHat Jr kids certs
SEO & Meta	4.5 / 10	Schema.org, OpenGraph, S	Missing robots.txt, sitemap.xml, and JSON-LD str
OVERALL COMPOSITE	6.1 / 10	Senior Staff Production	High-potential diamond in the rough requiring en

2. Prioritized Issue Register

[CRITICAL] Issues (Must Resolve Before Deployment)

1. Missing Resume PDF in Public Directory (HTTP 404)	CRITICAL
Problem: DeveloperVault.tsx links to /Resume.pdf, but the file does not exist in public/.	
Impact: Technical recruiters clicking 'Download PDF' receive a dead 404 error, breaking the primary user flow.	
Solution: Add an ATS-optimized, high-resolution Resume.pdf to /public/ and validate link integrity.	
2. Canvas Context Reset on Every Scroll Frame in ScrollyCanvas.tsx	CRITICAL
Problem: canvas.width and canvas.height are re-assigned inside the 60fps scroll callback.	
Impact: Purges hardware GPU raster cache and triggers continuous layout recalculations, causing frame drops and high CPU usage.	
Solution: Bind canvas sizing exclusively to debounced window resize events and apply window.devicePixelRatio for high-DPI.	
3. Uncontrolled 300-Image Network Burst on Page Mount	CRITICAL
Problem: 300 WebP frames (22 Megabytes) are loaded simultaneously via an unthrottled JavaScript loop.	
Impact: Saturates client connection pools, stalls critical CSS/font parsing, and triggers 300 status code errors.	
Solution: Load initial 15 keyframes immediately for instant playback, then stream remaining frames incrementally.	
4. Double / Triple Browser Tab Spawning on Project Clicks	CRITICAL
Problem: Nested onClick handlers in parent motion cards and child anchor links both execute window.open().	
Impact: Single click launches multiple redundant tabs and triggers Safari/Chrome popup blockers.	
Solution: Remove parent onClick triggers; rely strictly on semantic anchor links with stopPropagation().	

[HIGH] Priority Deficiencies

5. Zero HTTP Security Headers in Next.js Config	HIGH
Problem: next.config.mjs is blank; missing CSP, HSTS, X-Frame-Options, and Permissions-Policy.	
Impact: Vulnerable to clickjacking, unauthorized iframe embedding, and automated vulnerability scanners.	
Solution: Add headers() configuration in next.config.mjs defining strict security headers.	
6. Mock Reward API with No Email Delivery (/api/reward)	HIGH
Problem: /api/reward returns mock success without Resend/Postmark integration or database storage.	
Impact: Recruiter dossiers requested in Developer Vault are permanently lost; endpoint vulnerable to spam.	
Solution: Connect transactional email provider with Upstash Redis rate limiting and Zod schema validation.	
7. Global Cursor Destruction (WCAG 2.4.7 Violation)	HIGH
Problem: * { cursor: none !important; } wipes native pointers globally.	
Impact: Breaks accessibility for screen magnifiers and leaves user without pointer if JS encounter error.	
Solution: Scope custom cursor to fine-pointer media queries and restore native cursor on inputs and focus.	
8. Unconsented Audio Autoplay on General Click (WCAG 1.4.2)	HIGH
Problem: BackgroundAudio plays music on first page tap/click/keypress without explicit opt-in.	
Impact: Violates WCAG 2.2 Guideline 1.4.2; disruptive in work/office/meeting environments.	
Solution: Default audio to muted and require user interaction with the visualizer toggle.	
9. Continuous Off-Screen Animation Loop in GlobalMatrix.tsx	HIGH
Problem: D3 Geo render loop runs at 60 FPS continuously even when scrolled 4000px away.	
Impact: Wastes CPU/GPU compute cycles and drains mobile battery.	
Solution: Pause render loop via IntersectionObserver when component leaves viewport.	

3. Top 50 Production Improvements (Prioritized Directory)

#	CAT	REMEDIATION & IMPROVEMENT DIRECTIVE
01	ASSET	Add verified ATS-ready Resume.pdf to /public to fix 404 download error.
02	PERF	Stop resetting canvas width/height on every scroll frame in ScrollyCanvas.tsx.
03	PERF	Implement throttled chunk loading for 300 WebP sequence images.
04	BUG	Eliminate nested onClick handlers spawning multiple browser tabs on project launch.
05	SECURITY	Configure strict security headers in next.config.mjs (CSP, HSTS, X-Frame-Options).
06	SECURITY	Add rate limiting and Zod schema validation to /api/reward.
07	BACKEND	Implement real transactional email delivery for Developer Vault reward dossier.
08	A11Y	Remove global * { cursor: none !important } and restore native I-beam on inputs.
09	A11Y	Disable unconsented audio autoplay on general page tap/click.
10	PERF	Add IntersectionObserver to pause GlobalMatrix.tsx render loop when off-screen.
11	RECRUITER	Remove WhiteHat Jr certifications and replace with high-impact industry credentials.
12	RECRUITER	Clarify future dates on certifications (e.g. 'Candidate / In Progress').
13	SEO	Create src/app/robots.ts with crawler index directives.
14	SEO	Create src/app/sitemap.ts generating dynamic XML sitemaps.
15	SEO	Add comprehensive Schema.org Person & ProfilePage JSON-LD structured data.
16	BRAND	Unify all contact email references to contact@lakshya.uk.
17	PWA	Update site.webmanifest background and theme color to obsidian #07090e.
18	A11Y	Implement prefers-reduced-motion support across all Framer Motion components.
19	PERF	Apply window.devicePixelRatio scaling to Canvas backing buffers for Retina sharpness.
20	A11Y	Add aria-label and role='region' attributes to all interactive canvas containers.
21	UI/UX	Replace rapid oscillating sine wobble in Leadership.tsx with smooth inertial rotation.
22	ARCH	Unify RedGambit, Todar, and ElevateHub into a reusable compound <ProjectCard />.
23	CLEAN	Rename unbranded files in public/logos/ (images-2.jpeg, images.png).
24	CLEAN	Delete committed .DS_Store file from public/audio/.
25	PERF	Optimize BackgroundAudio to use preload='none'.
26	REFACTOR	Split monolithic Languages.tsx (981 lines) into modular sub-components and hooks.
27	REFACTOR	Split monolithic PixelFootballGame.tsx (923 lines) into game engine and HUD.
28	REFACTOR	Split Arcade.tsx (849 lines) into cartridge shelf, modal container, and state store.
29	ERROR	Add src/app/error.tsx and src/app/global-error.tsx React Error Boundaries.
30	ERROR	Add custom cyberpunk-styled src/app/not-found.tsx.
31	DESIGN	Unify background color tokens across globals.css, layout.tsx, and canvas.
32	DESIGN	Define reusable Tailwind utility tokens for glassmorphism panels and neon glows.
33	MOBILE	Refactor floating action buttons in bottom-right to stack ergonomically on 375px screens
34	MOBILE	Add touch-action: none to minigame canvases to prevent iOS bounce-scroll interference.
35	A11Y	Add keyboard navigation listeners (Tab, Space, Enter) to Languages weapon selection.
36	A11Y	Add aria-live='polite' to Terminal Hacker output buffer.
37	SEO	Add canonical link tag <link rel='canonical' href='https://lakshya.uk' />.
38	SEO	Add OpenGraph twitter handle and detailed description tags.
39	TYPESCRIPT	Eliminate all (as any) type assertions in GlobalMatrix.tsx.
40	PERF	Use dynamic import() for minigames to reduce initial JS payload.
41	PERF	Compress /public/projects/ JPG/PNG assets to WebP/AVIF formats.
42	PERF	Cache D3 Geo landmass calculations so TopoJSON isn't re-interpolated unnecessarily.
43	UX	Add visual tooltip explaining drag-to-aim controls in Pixel Football.
44	UX	Add quick sound mute/unmute keyboard shortcut (e.g. M key).
45	SECURITY	Replace inline dangerouslySetInnerHTML in layout.tsx with Next.js Script with nonce.
46	BUILD	Enable compress: true, poweredByHeader: false, and reactStrictMode: true.
47	DX	Add Prettier and ESLint rules enforcing strict import sorting and hook dependency checks
48	TESTING	Add Playwright end-to-end test suite checking all external links and canvas.
49	TESTING	Add Vitest unit tests for arcadeAudio procedural audio math and game physics.
50	DEVOPS	Configure GitHub Actions CI workflow to run linting, type-checking, and link validation.